# 1 Jarvis and Mckenzie, 1980 General equations

Conservation of mass, momentum and energy:

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$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0 \tag{1}$$

$$\rho \left( \frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} \right) = -\nabla P + \rho g + \frac{\partial}{\partial x_j} \tau_{ij} \tag{2}$$

$$\rho C_p \left\{ \frac{\partial T}{\partial t} - \frac{\alpha T}{\rho C_p} \frac{\partial P}{\partial t} + \boldsymbol{u} \cdot [\nabla T - (\nabla T)_s] \right\} = \nabla \cdot (K \nabla T) + H + \tau_{ij} \frac{\partial u_i}{\partial u_{ij}} \tag{3}$$

All terms and symbols used in these equations are shown in table 2. The (Newtonian) relation between the deviatoric stress tensor $\tau_{ij}$ and the velocity field is

$$\tau_{ij} = \eta \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \delta_{ij} \frac{\partial u_k}{\partial x_k} \right) + \eta_2 \delta_{ij} \frac{\partial u_k}{\partial x_k}. \tag{4}$$

Combining Eq. (4) with the momentum equation (2) gives the general Navier-Stokes equations, but Jarvis and Mckenzie (1980) does not show these. Instead, they assume that $\frac{1}{Pr} \to 0$, which they assume (do they?) is the case for the Earth's mantle, the inertial forces can be ignored. The resulting equations will be discussed in the next section, which covers the approximations made by Jarvis and Mckenzie (1980). We also define the constant Grüneisen's parameter:

$$\Gamma = \frac{\alpha K_s}{\rho C_p} = \frac{\alpha K_T}{\rho C_v} \tag{5}$$

Where $K_s$ and $K_T$ are the isentropic and isothermal moduli of bulk compressibility, and $C_v$ and $C_p$ are the specific heat at constant volume and pressure, respectively (2).

Additional equations used in derivations in the next section:

$$C_p - C_v = \frac{\alpha^2 K_T T}{\rho} \tag{6}$$

# 2 Jarvis and Mckenzie, 1980 Approximations

Problem statement: for the Boussinesq approximation to be valid, the depth range of convection $d$ must be small compared to the temperature scale height $H_T$ ($d/H_T \ll 1$). This is not the case for many systems of deep convection, amongst which the convection of planetary mantles. Physically, $d/H_T \approx O(1)$ means that the influence of viscous dissipation, variable adiabatic temperature- and density gradients must be maintained in the set of governing equations.
The assumptions made for the **anelastic liquid approximation** are that $d = O(H_T)$ and that $\alpha T \ll 1$ what does this mean and how does this relate to Prandtl number? and if $\alpha \ll 1$, how come $D$ is not $\ll 1$?

## 2.1 Approximate momentum equation

Since volume changes in the Earth's mantle occur over timescales greater than $10^7$ years, terms containing the bulk viscosity $\eta_2$ are neglected. Substituting from Eq. (4), the momentum equation, Eq. (2), becomes

$$0 = -\nabla P + \rho \mathbf{g} + \frac{\partial}{\partial x_i} \eta \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \nabla \eta \, (\nabla \cdot \mathbf{u}).$$

## 2.2 Reference state and equation of state

A reference state with hydrostatic pressure $P_H$ and adiabatic temperature distribution (gradient?) $T_s$ is considered. The equation of state is written as a linear Taylor expansion of $\rho$ about the reference state density distribution $\rho_r$:

$$\rho(T, P) = \rho_r [1 - \alpha (T - T_s) + K_T^{-1} (P - P_H)] \tag{7}$$

Where

$$\alpha = \frac{1}{\rho_r}\left(\frac{\partial \rho_R}{\partial T}\right)_P \tag{8}$$

$$K_T = \rho_r \left(\frac{\partial P}{\partial \rho_r}\right)_T \tag{9}$$

Determining the reference density distribution $\rho_r$ is done by considering the gradient of $\rho_r(S, P_H)$ with $S$ the entropy per unit mass:

$$\nabla \rho_R(S, P_H) = \left(\frac{\partial \rho_r}{\partial S}\right)_{P_H} \nabla S + \left(\frac{\partial \rho_r}{\partial P_H}\right)_S \nabla P_H \tag{10}$$

With $\nabla S = 0$ since the reference state is adiabatic, and with $\nabla P_H = -\rho_r g \hat{\boldsymbol{z}}$:

$$\frac{1}{\rho_r}\frac{\partial \rho_r}{\partial z} = -\frac{\rho_r g}{K_s} \tag{11}$$

with $K_s = \rho_r \left(\frac{\partial P_H}{\partial \rho_r}\right)_s$. Using the definition of the Grüneisen parameter $\Gamma$ and the temperature scale height of the liquid $H_T$ gives the Adams-Williamson relation add images!

$$\frac{1}{\rho_r}\frac{\partial \rho_r}{\partial z} = -\frac{1}{H_T \Gamma} \tag{12}$$

Which is valid for a **chemically homogeneous adiabatic density distribution under hydrostatic pressure**. Integration of equation 12 over $z$ gives the reference densitiy distribution as

$$\rho_r(z) = \rho_0 \exp[(d - z)/H_T \Gamma] \tag{13}$$

where $\rho_0 = \rho_r(z = d)$ is the reference density value at the surface orientation of coordinate system?? Plugging this expression of the reference density distribution into 7 gives the (linearized equation) of state

$$\rho(T, P) = \rho_0 \exp[(d - z)/H_T \Gamma][1 - \alpha(T - T_s) + K_T^{-1}(P - P_H)] \tag{14}$$

## 2.3 Nondimensionalization

3, 3. move tables here? In this section, Jarvis and Mckenzie, 1980 introduces various characteristic values of variables in order to nondimensionalize the preceding equations. These characteristic values and the resulting dimensionless variables are presented in the tables below.The characteristic value for **velocity** is taken directly from McKenzie et al., 1974, and their derivation is expanded below:

Our starting point is the momentum equation (in the notation of McKenzie et al., 1974), with the left-hand side set to 0 (on the basis of $\mathrm{Re} \ll 1$ and $\frac{1}{\mathrm{Pr}} \to 0$):

$$0 = -\frac{1}{\rho}\frac{\partial P}{\partial x_i} + \frac{\partial \phi}{\partial x_i} + \frac{1}{\rho}\frac{\partial \tau_{ij}}{\partial x_j} \tag{15}$$

with

$$\tau_{ij} = \rho\nu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$

$$\nabla \phi = \boldsymbol{g}$$

last term in Eq. (15) becomes help index confusion

$$\frac{1}{\rho}\frac{\partial \tau_{ij}}{\partial x_j} = \rho\nu\left(\frac{\partial}{\partial x_j}\frac{\partial u_i}{\partial x_j} + \frac{\partial}{\partial x_j}\frac{\partial u_j}{\partial x_i}\right)$$

$$= \rho\nu(\nabla^2 u_i + \frac{\partial}{\partial x_i}\nabla \cdot \boldsymbol{u})$$

**assuming incompressibility,** $\nabla \cdot \boldsymbol{u} = 0$, and or should this be $\boldsymbol{u}$ (vector)?

$$\frac{1}{\rho}\frac{\partial \tau_{ij}}{\partial x_j} = \nu \nabla^2 u_i \tag{16}$$

Now we formulate an equation of state under the Boussinesq approximation (Taylor expansion of density $\rho(T)$ around a reference state temperature $T_0$):

$$\begin{aligned}
\rho &= \rho_0[1 - \alpha(T - T_0)] \\
\frac{1}{\rho} &= \rho_0^{-1}[1 - \alpha(T - T_0)]^{-1} \\
&\approx \rho_0^{-1}[1 + \alpha(T - T_0)]
\end{aligned}$$

with $\alpha = -\frac{1}{\rho}\left(\frac{\partial \rho}{\partial T}\right)_P$ the thermal expansivity coefficient.

We can split the pressure gradient $\nabla P$ into a hydrostatic part ($\nabla P_s$) and a non-hydrostatic part $\nabla p$:

$$\begin{aligned}
\nabla P &= \nabla P_s + \nabla p \\
&= \rho_0 \boldsymbol{g} + \nabla p
\end{aligned}$$

Together with the equation of state, the pressure term + gravity in Eq. (15) then becomes

$$\begin{aligned}
-\frac{1}{\rho}\nabla P + \boldsymbol{g} &= -\rho_0^{-1}(1 + \alpha(T - T_0))(\nabla P_s + \nabla p) + \boldsymbol{g} \\
&= -\rho_0^{-1}\nabla P_s - \rho_0^{-1}\nabla P_s \alpha(T - T_0) - \rho_0^{-1}\nabla p - \rho_0^{-1}\nabla p \alpha(T - T_0) + \boldsymbol{g} \\
&\approx -\boldsymbol{g}\alpha(T - T_0) - \rho_0^{-1}\nabla p
\end{aligned}$$

Where we neglected the second-order terms ($\sim \nabla p T$) and used that $\nabla P_s = \rho_0 \boldsymbol{g}$, cancelling out the gravity terms. With this, Eq. (15) becomes

$$\nu \nabla^2 \boldsymbol{u} = \boldsymbol{g}\alpha(T - T_0) + \rho_0^{-1}\nabla p \tag{17}$$

**note that $\nabla p$ is only the non-hydrostatic pressure gradient**.

We can take the curl of Eq. (17), using that $\nabla \times \boldsymbol{u} = \boldsymbol{\omega}$, that the curl of a gradient is always zero ($\nabla \times (\nabla f) = 0$), and that $\boldsymbol{g} = \nabla \phi$:

$$\begin{aligned}
\nu \nabla^2 \boldsymbol{\omega} &= \nabla \times (\boldsymbol{g}\alpha(T - T_0)) + \nabla \times (\rho_0^{-1}\nabla p) \\
&= \nabla \times [\boldsymbol{g}\alpha(T - T_0)] \\
&= \alpha(T - T_0)(\nabla \times \boldsymbol{g}) + \alpha \nabla T \times \boldsymbol{g} \\
&= \alpha \nabla T \times \boldsymbol{g}
\end{aligned} \tag{18}$$

Now we assume that the flow in restricted to the $x, z$ plane, meaning $\boldsymbol{u} = (u, 0, w) = \left(-\frac{\partial \psi}{\partial z}, 0, \frac{\partial \psi}{\partial x}\right)$ (where $\psi$ is the stream function) and thus $\boldsymbol{\omega} = (0, \omega, 0) = (0, -\nabla^2 \psi, 0)$. With $\boldsymbol{g} = (0, 0, g)$, Eq. 18 becomes (computing the cross product)

$$\nabla^2 \omega = \frac{\alpha g}{\nu}\frac{\partial T}{\partial x} \tag{19}$$

From this equation, we can get an characteristic velocity for the convecting system.

$$\begin{aligned} \mathcal{O}(\nabla^2\omega) &= \mathcal{O}(\frac{\alpha g}{\nu}\frac{\partial T}{\partial x}) \\ \frac{U}{d^3} &= \frac{\alpha g}{\nu_0}\frac{T_1}{d} \\ U &= \frac{\alpha g T_1 d^2}{\nu_0} \end{aligned} \tag{20}$$

Where $U$, $\nu_0$, and $T_1$ are characteristic values for velocity, kinematic viscosity, and temperature deviations from the adiabatic reference profile $T_0$, respectively.

For the scaling of pressure variations $P_1$ are derived by considering the horizontal component of Eq. (**??**) and assuming the non- hydrostatic pressure gradient $\nabla P$ and the viscous force terms to be of the same order of magnitude is this an illegal way to write this?:

$$\begin{aligned} \mathcal{O}(\nabla P) &= \mathcal{O}\left(\frac{\partial}{\partial x_i}\eta\left(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}\right)-\frac{2}{3}\nabla\eta\left(\nabla\cdot\mathbf{u}\right)\right) \\ \frac{P_{1_r}}{d} &= \frac{\eta U}{d^2} \\ P_{1_r} &= \frac{\eta U}{d} \end{aligned}$$

Thus, for the dimensionless pressure variable, using Eq. (20) and the kinematic viscosity $\nu_0 = \frac{\eta}{\rho_0}$, we get:

$$\begin{aligned} P_1' &= \frac{P_1}{P_{1_r}} = \frac{P_1 d}{\eta U} \\ &= \frac{P_1}{\alpha T_1 \rho_0 g d} \end{aligned}$$

For the dimensionless energy generation rate $\epsilon_0 = \frac{H}{\rho C_p}$ we consider the heat generation and the time scale in the energy equation (3): better way to write this down? order of magnitude way?

$$\begin{aligned} \frac{\partial T}{\partial t} &\sim \epsilon_0 \\ \frac{T_1}{\tau}\frac{\partial T'}{\partial t'} &\sim \epsilon_0 \\ \epsilon_0' &= \frac{\epsilon_0 \tau}{T_1} \\ &= \frac{\epsilon_0 \nu_0}{T_1^2 g \alpha d} \end{aligned}$$

Where expressions from table 3 were used to arrive at the final expression. Similarly for the thermal diffusivity, we balance the conduction term with the time-derivative term in the energy equation (3) :

$$\begin{aligned} \mathcal{O}(\kappa_0\nabla^2 T) &= \mathcal{O}\left(\frac{\partial T}{\partial t}\right) \\ \kappa_{0,r}\frac{T_1}{d^2} &= \frac{T_1}{\tau} \\ \kappa_{0,r} &= \frac{d^2}{\tau} \\ \kappa_0' &= \frac{\kappa_0}{\kappa_{0,r}} \\ &= \frac{\kappa_0 \nu_0}{d^3 g \alpha T_1} \end{aligned}$$

Finally, for the dimensionless boxwidth, we simply divide the dimensional box width with the characteristic length:

$$\lambda = \frac{L}{d}$$

Table 1: characteristic values of Jarvis and Mckenzie, 1980

| Variable | Characteristic value | Dimensionless variable |
|---|---|---|
| depth | $d$ | $(x', z') = \dfrac{(x, z)}{d}$ |
| temperature | $T_1 (= 1\text{K})$ | $T' = \dfrac{(T - T_0)}{T_1}$ |
| density | $\rho_0$ | $\rho_r' = \dfrac{\rho_r}{\rho_0}, \quad \rho = \dfrac{\rho}{\rho_0}$ |
| velocity | $U = \dfrac{g\alpha T_1 d^2}{\nu_0}$ | |
| kinematic viscosity | $\nu_0 = \eta/\rho_0$ | |
| time | $\tau = d/U$ | |

## 2.4 The anelastic-liquid approximation

From now on, only the dimensionless variables will be used but with primes surpressed **(i.e. all variables are assumed dimensionless)**. The dimensionless equation of state (from eq. 14) becomes

$$\rho = \rho_r[1 - \mu(T - T_s) + \mu(\rho_0 g d/K_T)P_1] \tag{21}$$

with $\mu = \alpha T_1$. If $\mu \to 0$, $\rho$ reduces to its reference profile $\rho_r$ (as is the case in the Boussinesq approximation).

Now, we want to eliminate the bulk modulus from Eq. (21). For this, we use the Grüneisen parameter (5) and the specific heat identity (6) to rewrite the last term:

$$\begin{aligned}\frac{\rho_0 g d}{K_T} &= \frac{\rho_0 g d}{\Gamma \rho C_p}\alpha(1 + \alpha\Gamma T) \\ &= \frac{\rho_0}{\rho}\frac{d}{H_T\Gamma}(1 + \alpha\Gamma T) \\ &= \frac{D}{\rho_r\Gamma}(1 + \alpha\Gamma T)\end{aligned}$$

Where it was used that $\rho \to \rho_r$ and primes were dropped, and that $d/H_T = D$.

Now we assume $T \approx T_s$ and Jarvis and Mckenzie (1980) (implicitly) introduces the adiabatic temperature profile: NOT SURE ABOUT THIS

$$T_s = T_0 \exp\{(d - z)/H_T\} = T_0 \exp\{(1 - z)D\} \tag{22}$$

Where we used that $z' = z/d$ and dropped the prime. Subsituting (22) into (21) gives WHY WHY WHY derive this equation if you're just going to throw away all terms with $\mu$???

$$\rho = \rho_r \left\{1 - \mu(T - T_s) + \mu\left(\frac{D}{\rho_r\Gamma}\right)[1 + \mu\Gamma T_0 \exp\{(1 - z)D\}]P_1\right\} \tag{23}$$

**Now, some approximations are made:**

- First, the Mach number $M^2$ is evaluated. This dimensionless number measures the speed of the fluid relative to the local speed of sound. Using the characteristic velocity $U$ derived in section 2.3 (Table 3) and the velocity of sound $C = \sqrt{\frac{1.8K_s}{\rho}} = \sqrt{\frac{1.8C_p\Gamma}{\alpha}}$ where does the 1.8 come from??

$$\begin{aligned}M^2 &= \frac{U^2}{C^2} \\ &= \frac{d}{H_T}\alpha T_1 \frac{g\alpha T_1 d^3}{1.8\nu_0^2\Gamma} \\ &= \frac{d}{H_T}\mu R * Pr^{-1}\frac{1}{1.8\Gamma}\end{aligned}$$

Now, for $\mathrm{O}(\Gamma) = 1$, $\mathrm{O}(\frac{d}{H_T})$, and $Pr \to \inf$, the Mach number $M^2 \ll 1$. Based on this approximation, $\frac{\partial \rho}{\partial t}$ is neglected form the mass continuity equation (1).

- Since $\alpha \approx 10^{-5}\mathrm{K}^{-1}$ and $T_1 = 1\mathrm{K}$, $\mu \ll 1$ and from Eq. (23) we can see that $\rho = \rho_r$ best way to format units in math mode?

Now, we can plug in the dimensionless $\rho_r$ (13) into the mass continuity equation (1):

$$\begin{aligned} \nabla \cdot [\mathbf{u} \exp((\mathbf{1} - \mathbf{z})\mathbf{D}/\mathbf{\Gamma})] = 0 \\ \nabla \cdot [\mathbf{u} \exp(-\mathbf{z}\mathbf{D}/\mathbf{\Gamma})] = 0 \end{aligned} \tag{24}$$

Since $\Gamma$ and $D$ are both constant. check D to be constant to be sure. also why are they bold? and is D italic/plain?

so, for own sanity, a summary: the density in the mass continuity equation is reduced to the reference profile (neglecting variations away from density profile) while the full density (including variations) are used in the momentum equation (where density is multiplied with gravity). In the boussinesq equation, is the density also reduced in the body force term?

**Restricting the flow to the $x$,$z$ plane** and using that $\mathbf{u} \exp(-\mathbf{z}\mathbf{D}/\mathbf{\Gamma})$ is divergence-free (solenoidal), a stream function can be defined:

# 3 Appendix 1: tables of symbols

Table 2: Symbols used by Jarvis and Mckenzie, 1980

| | | |
|---|---|---|
| $d$ | | depth of the convecting layer |
| $H_T$ | $\frac{C_p}{g\alpha}$ | temperature scale height of the liquid |
| $D$ | $\frac{d}{H_T}$ | |
| $C_p$ | | specific heat at constant pressure |
| $C_v$ | | specific heat at constant volume |
| $\alpha$ | | volume coefficient of thermal expansion |
| $\tau_{ij}$ | | deviatoric stress tensor |
| $(\nabla T)_s$ | $\frac{\alpha T}{\rho C_p}\nabla P$ | adiabatic temperature gradient |
| $K$ | | thermal conductivity |
| $H$ | | rate of internal heat production |
| $\eta$ | $\rho\nu$ | dynamic viscosity |
| $\nu$ | | kinematic viscosity |
| $\eta_0$ | $\ll 1$ | second or bulk viscosity |
| $\Gamma$ | $\frac{\alpha K_s}{\rho C_p} = \frac{\alpha K_T}{\rho C_v}$ | Grüneisen's parameter, assumed constant and $O(1)$ |
| $K_s$ | | isentropic modulus of bulk compressibility |
| $K_T$ | | isothermal modulus of bulk compressibility |
| $\mu$ | $\alpha T_1$ | |
| $\alpha$ | $10^{-5} \quad {}^\circ K^{-1}$ | volume coefficient of thermal expansion |

Table 3: characteristic values of Jarvis and Mckenzie, 1980

| | |
|---|---|
| depth | $d$ |
| temperature | $T_1 = \frac{1}{\mathrm{K}} WHAT?? \rightarrow 1^\circ\mathrm{K}$ |
| density | $\rho_0$ |
| velocity | $U = \frac{g\alpha T_1 d^2}{\nu_0}, \quad \nu_0 = \eta/\rho_0$help this is different from schubert |
| time | $\tau = d/U$ |

Table 4: Dimensionless variables and numbers in Jarvis and Mckenzie, 1980

| | |
|---|---|
| $(x', z')$ | $(x, z)/d$ |
| $\rho'_r$ | $\rho_r/\rho_0 = \exp[(1 - z')D/\Gamma], \quad D = d/H_T$ |
| $\rho'$ | $\rho/\rho_0$ |
| $T'$ | $(T - T_0)/T_1$ |
| $t'$ | $t/\tau$ |
| $P'_1$ | $\dfrac{P_1 d}{U\eta} = P_1 \alpha T_1 \rho_0 g d, \quad P_1 = P - P_H$ |
| $\kappa'_0$ | $\kappa_0\nu_0/g\alpha T_1 d^3$, thermal diffusivity |
| $\epsilon'_0$ | $\epsilon_0\nu_0/g\alpha T_1^2 d$, energy generation rate, $\epsilon_0 = H/\rho_0 C_p$ |
| $\lambda$ | $L/d$, box width, $L$ is the dimensional box width |
| $M^2$ | $U^2/C^2$ Mach number, where $C$ is sound speed |
| $R*$ | $1/\kappa'_0$ Rayleigh number equivalent |
| $Pr$ | $\nu'_0/\kappa'_0$ Prandtl number |